iLearn Practice Exam

10BR · Chapter / Topic 2

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 2: Data Transmission

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 75

Marks obtained: ..... / 75

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 9 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 Complete the statements about data packets and packet switching.

Use the terms from the list.

Some of the terms in the list will **not** be used. You should only use a term once.

destination address first footer header

last main data packets payload

routers servers trailer

Data is broken down into packets. A data packet has a packet ..... that contains the packet number and the ..... .

Each packet could take a different path from the sender to the receiver; this is controlled by ..... .

Packets may arrive out of order. Once the ..... packet has arrived, the packets are reordered.

[4]

2 Data is transmitted using the internet.

(a) The data is encrypted before transmission.

(i) State the purpose of encrypting data.

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..... [1]

(ii) Describe the differences between symmetric and asymmetric encryption.

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..... [4]

(b) The data is transmitted using packet switching.

(i) Identify **two** items of data contained in a packet header.

1 .....

2 .....
[2]

(ii) Give the purpose of a router in the packet-switching process.

.....

..... [1]

3 Data packets are transmitted across a network from one computer to another computer.

(a) Describe the structure of a data packet.

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..... [3]

(b) Packet switching is used to transmit the data packets across the network.

Identify the device that controls which path is taken by each data packet.

..... [1]

(c) Serial data transmission is used to transmit the data packets across the network.

Explain why serial data transmission is used to transmit the data packets.

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..... [3]

4 The table contains statements about error detection methods.

Complete the table by giving the correct error detection method for each statement.

error detection method	statement
.....	An odd or even process can be used.
.....	A value is calculated from the data, using an algorithm. This happens before and after the data is transmitted.
.....	A copy of the data is sent back to the sender by the receiver.
.....	Acknowledgement and timeout are used.
.....	A value is appended to data that has been calculated using the data. This value is checked on data entry.

[5]

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5 A company decides to create a network for its devices.

All the company devices are within a single room.

The employees will need to use their devices to send data to each other and share files.

The company decides to send data across the network using packet switching.

(a) An employee sends an email to another employee. The email is broken down into packets.

Describe the structure of a packet of data for the email.

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..... [4]

(b) The company decides to use parallel full-duplex data transmission to send the data across the network.

(i) Explain the reasons why the company have chosen this method of data transmission.

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..... [4]

(ii) Give **two** drawbacks of the company using this method of data transmission.

1 .....

.....

2 .....

.....

[2]

(iii) Give **one** other method of data transmission the company could have chosen.

..... [1]

6 A restaurant has an electronic ordering system. The system allows customers to select the food they want to order. The data for the order is transmitted across their network to a computer in the restaurant kitchen.

(a) The data is sent using serial full-duplex data transmission.

(i) Give **two** reasons why serial full-duplex is a suitable data transmission method for the data.

1 .....

.....

2 .....

..... [2]

(ii) Give **one** improvement that would be made to the data transmission if parallel transmission is used instead of serial transmission.

.....

..... [1]

(b) A checksum is used to check the data for errors after transmission.

(i) Describe the process of the checksum error detection method.

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..... [5]

(ii) Identify **one** other error detection method that could be used to check for errors in the data.

..... [1]

(c) A customer can also order food from home using the internet. The data for their order is broken down into packets and transmitted across the internet to the computer in the restaurant kitchen.

(i) Tick (✓) **one** box to show which item would **not** be included in a packet's header.

A destination address ☐

B originator's address ☐

C packet number ☐

D payload ☐

[1]

(ii) The packets of data may need to be reordered when they arrive at the computer in the restaurant kitchen.

Explain why the packets of data may need to be reordered.

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..... [2]

7 Data is sent from an employee's computer to a server. The data is correct when it is sent, but an error is detected when the data is received.

(a) Describe how an error can occur in the data during transmission.

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.....

.....

..... [2]

(b) Methods of error detection and correction can be added to make sure that errors are identified after data is transmitted.

Complete the statements about methods of error detection and correction.

Use terms from the list. **Not** all terms need to be used. You should only use each term once.

acknowledgement automatic repeat query (ARQ) binary bit

byte calculation check digit checksum correct

echo check error even incorrect negative

number odd positive timeout

A parity check is set to be ..... or

..... . A parity ..... is

added to each byte before transmission to make it match the set parity.

An ..... involves comparing the data sent to the data

received back from the receiving device.

An ..... uses acknowledgement and

..... . The acknowledgement system can be

..... or ..... .

[8]

8 Data is split into packets before it is transmitted across a network.

(a) The diagram shows a packet with **three** sections.

Complete the diagram by writing the name of each section of the packet.

Write an example in each section of what would be stored in that part of the packet.

Packet

The first section is called the ..... →

First
Example .....

The second section is called the ..... →

Second
Example .....

The third section is called the ..... →

Third
Example .....

[6]

(b) Packets can be sent across a network using different methods of data transmission.

Explain how data is transmitted in the **two** given methods of data transmission.

(i) Serial full-duplex

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..... [4]

(ii) Parallel half-duplex

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..... [4]

9 An optical mouse is connected to a computer using a universal serial bus (USB) cable.

(a) The USB cable transmits data using the serial half-duplex data transmission method.

Describe how the serial half-duplex data transmission method transmits the data.

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..... [4]

END OF QUESTION PAPER